Exploring and Evaluating Multisource Remote Sensing for Poverty Mapping in Mexico: Integrating Geomedian and VIIRS Night-Time Light Data

Allan Francisco Ramírez-González, Marivel Zea-Ortiz, Pablo Vera-Alfaro, and Delfino Vargas-Chanes

Abstract In recent years, the use of remote sensing data combined with deep learning has become increasingly popular for mapping socioeconomic indicators, benefiting from the growing availability of large-scale Earth observation datasets. This study examines the integration of remote sensing sources with varying numbers of spectral bands and resolutions to enhance the inference capabilities of convolutional neural networks (CNNs) for predicting multidimensional poverty in Mexico. We explore and evaluate two methods for combining daytime satellite imagery and nighttime lights to predict multidimensional poverty at the census track level (Basic Geostatistical Area, AGEB) in Mexico. To achieve this, we gather Landsat-8 imagery (comprising 6 spectral bands) and VIIRS annual composite nighttime lights (NTL) and combine them into a single numeric array in two distinct ways. The first method consists of a band composed entirely of a single NTL intensity AGEB statistic value correlated with the poverty rate. The second method integrates rescaled NTL pixel values as a seventh input band, spatially aligned with the Geomedian imagery. To see if proposed methods improve poverty inference, we train five CNN regression models using the 2020 Municipal Multidimensional Poverty measure from CONEVAL as reference values: (1) using Geomedian only, (2) Geomedian plus derived NTL value, (3) Geomedian plus rescaled NTL by Nearest Neighbor algorithm while (4) and (5) using same method but resampled by Bilinear and Cubic kernels. Predictions are generated at the AGEB level and aggregated to the municipal and state levels using weighted averages. The results show that the seven-band models outperform the six-band model. The best-performing model achieved an R^2 of 0.76 at the municipal level, and up to 0.96 at

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the state level. Our research demonstrates that combining multi-source satellite data, including those with different spatial and spectral resolutions, can improve CNN performance for poverty prediction at different aggregation levels. This approach supports more accurate and granular socioeconomic analysis, contributing to data-driven public policy efforts in Mexico.

Keywords Poverty prediction · Satellite imagery · Deep learning

1 Introduction

In countries such as Mexico and Latin America, in general, it is essential to generate information related to poverty, as it is a fundamental part of developing social policies that enable the fulfillment of one of the Sustainable Development Goals (SDGs) established by the United Nations Development Programme, which is the eradication of poverty. This project aims to explore alternative methods for measuring poverty using census data, multispectral images from Landsat, nighttime images from the Visible Infrared Imaging Radiometer Suite (VIIRS), and deep learning.

Providing timely information is crucial for the development of social policy. Official surveys are costly, their coverage is limited, and the frequency with which information is obtained is very long (varying between every 2, 5, or 10 years). The proposed project seeks to minimize these costs and obtain reliable indicators with the desired frequency at the subnational level. With this information, research can be conducted more frequently and made available to decision makers.

In this study, we explore and evaluate two methods for combining daytime satellite imagery and nighttime lights to predict multidimensional poverty at the AGEB level (Basic Geostatistical Area, a subdivision of municipal geostatistical areas in Mexico, equivalent to a census track) as defined by INEGI (The National Institute of Statistics and Geography) [1] in order to generate inferences at municipal and state level. To achieve this, we gather Landsat-8 imagery and VIIRS annual composite nighttime lights and combine them into a single numeric array in two distinct ways.

The CNN models were trained using satellite information available at the AGEB level and poverty data for 2020, as reported by CONEVAL (Consejo Nacional de Evaluación de Política Pública) [2], the Mexican agency responsible for measuring poverty in Mexico. Our main contribution is to combine both remote sensing resources in order to improve poverty prediction through the use of CNN. With the models obtained, it will be possible to estimate poverty not only for the years when information is available, but also to test their usefulness for subsequent years, even when complete poverty information is lacking, but with satellite data available.

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2 Related Work

The use of high- and low-resolution satellite imagery is a growing trend in the mapping of various socioeconomic indicators. Many studies, such as those by [3–12], have incorporated nighttime data from various satellite sensors to predict poverty values at a granular level. For many researchers, this is the most cost-effective way to map poverty when no surveys or censuses are scheduled for that year. Research linking NTL to economic well-being and poverty has been highly successful since [13].

Studies combining NTL with other remote sensing inputs have gained popularity due to their spatial and temporal compatibility, as well as the favorable results they have yielded. NTL has been integrated in different ways. The first combination of daytime and nighttime approaches for poverty prediction was proposed by [4, 14] where a CNN was trained with daytime images to predict the intensities of NTL since NTL is highly correlated with economic activity [13, 15, 16]. Additional investigations used NTL intensities as a class label to train a classification model and subsequently predict continuous poverty rates [5, 8].

Other approaches in addition to CNN have been explored. For example [6, 7, 17], combined NTL with a normalized vegetation index (NDVI), built-up index (BUI), normalized water index (NDWI), land surface temperature (LST), carbon monoxide (CO), nitrogen dioxide (NO_2) and Point of Interest data (POI) from Open Street Map (OSM) as zonal statistics in order to feed multiple ML models incorporating Principal Component Analysis (PCA), Support Vector Regression (SVR), Decision Tree Regresion (DTR) and Random Forest Regression (RFR) as well as a one-dimensional CNN (1D-CNN).

NTL combination approaches are not limited between nighttime satellites. Xu et al. [9] and Yong et al. [11] combined VIIRS and DMSP using multiple images to create a large NTL time series data and run diverse regression models such as linear regression, SVR, RFR, and Gaussian Process Regression (GPR). Most recently Yeh et al. [10] introduced a Double Branch CNN approach where two separate CNN architectures are trained, one with Landsat and another using NTL and the final layers of both models are fused. It was also used in [3, 18] and showed promising results in different African countries (Table 1).

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In our study, we use the municipal values of the CONEVAL poverty indicator measured in 2020 as reference values. Therefore, throughout this article, the term poverty should be understood as referring to multidimensional poverty. We train CNN regression models to predict the CONEVAL poverty indicator from features learned from both daylight and nighttime satellite data. In our method, we combine both types of data for each region into a single image, which we use as input to the CNN. The models can thus make inferences in a single step.

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Table 1 Studies combining NTL with other remote sensing resources

Authors	Algorithm	Remote sensing	NTL integration	Analysis level	Indicator	Results
Abbes [3], Yeh [10], Jean [4], Tingzon [8], Xie [14]	CNN	Landsat 5-8, VIIRS, DMSP, Google Static Maps, OSM	CNN with NTL as label (classification or regression)	Village, cluster, (household groups)	Wealth index, asset index, consumption, access to services	$R^2 : 0.55 - 0.84$, accuracy: 0.71
Xu [9], Yong [11]	Regression (multiple, nonlinear, decision trees)	DMSP, VIIRS (reprocessed, combined and rescaled)	NTL as an explanatory variable in regression models	County	Poverty incidence, (poverty index)	$R^2 : 0.60 - 0.83$[a]
Putri [6]	PCA	Sentinel-2, VIIRS, MODIS, Sentilen-5P, OSM	NTL as a weighted variable	Subdistrict	Relative spatial poverty index	$R^2 : 0.50$
Putri [7]	SVR, RFR, MLP, CNN-1D	Sentinel-2, MODIS, Sentinel-5P, VIIRS, OSM	NTL classified into 3 classes as a label	District and subdistrict	Poverty rate	$R^2 : 0.84$ (CNN-1D, district level)

[a] Only verified on the south China Region

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3 Methodology

We trained various CNN models based on the EfficientNet-B3 architecture [19] and applied transfer learning with weights learnt by "ImageNet" [20] which has been proven to be an effective way of training deep learning models for poverty prediction [4, 14]. It was also seen that CNN-based models showed better results than other ML algorithms, many investigations used ResNet34, VGG (F and 16 versions) [3, 4, 7, 8]. However, as demonstrated in previous work [21], EfficientNet-B3 obtained great results.

3.1 Data Collection

The objective of our model is to forecast poverty rates at the municipal, state, and national levels. In Mexico, official poverty statistics at the AGEB level are currently unavailable. Nevertheless, to construct our poverty prediction model, it was necessary to allocate a poverty value to each AGEB unit. To achieve this, the 2020 Municipal Multidimensional Poverty Estimation data set provided by CONEVAL was employed as a proxy variable.

CONEVAL currently offers poverty estimates at the municipal level, but these statistics are only accessible for the years 2010, 2015, and 2020 —years aligned with census and ENIGH (National Household Income and Expenditure Survey) data collection. For other years, municipal-level data is unavailable. In such instances, CNN methods could be implemented to estimate poverty in the absence of direct data.

In Mexico, official poverty statistics at the AGEB level are unavailable. Consequently, the 2020 Municipal Multidimensional Poverty Estimate, published by CONEVAL, was employed as a proxy variable for training our CNN model, substituting for AGEB-specific data. This methodology presupposes that poverty is relatively uniform across each municipality and this estimate can serve as an approximation for all subunits. While this simplification may introduce bias due to heterogeneity within municipalities, it facilitates the application of supervised learning techniques at finer spatial resolutions using publicly accessible official data.

As daytime imagery, we use Landsat-8 Geomedian 2020 for Mexico [22], composed of 144 mosaics of 150 km $\times$ 150 km each and contains six spectral bands: Red, Green, Blue (RGB), short-wave infrared 1 (SWIR1), short-wave infrared 2 (SWIR2), and near-infrared (NIR), all with a pixel resolution of 30 m. This imagery is obtained from the Geospatial Data Cube of Mexico by INEGI. The NTL data are obtained from the Visible Infrared Imaging Radiometer Suite Day/Night Band (VIIRS DNB), and we used the 2020 Mexico annual composite, which can be downloaded from NASA LAADS DAC suite. The pixel values on the snow-free layer vary from 0 to 4923 and have 15 arc-second/pixel resolution, which is approximately 450 m.

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We use 40 × 40 pixel patches based on the 2020 Geostatistical Framework developed by the National Institute of Statistics and Geography (INEGI) [1], which contains the coordinates and polygons of the 81428 AGEB, urban and rural, in the national territory. We excluded 23 AGEBs corresponding to three municipalities for which CONEVAL did not report a poverty measure. To generate the centroid of each AGEB, we used QGIS and extracted the corresponding multispectral image patches (geomedian) for each one. During the training of the CNN model, we rotate each image patch at a random angle and perform additional cropping to obtain a final size of 25 × 25 pixels, corresponding to a resolution of 750 × 750 m.

3.2 Combining VIIRS Data with LANDSAT Geomedian

Our approach integrates both daytime imagery and NTL intensity into a single numeric array in two distinct ways. First, we modified the first convolutional layer of the EfficientNet-B3 CNN architecture to accept an input with seven channels instead of the default 3. The rest of the network remains unchanged. Then we resized the input images to 224 × 224 pixels, which matches the expected input dimensions of the model.

As demonstrated in [23], several statistical features of the intensity of NTL, including the log of the median, log min, median, are highly negative correlated with poverty at the municipal level. We obtained these statistics at the AGEB level having results similar to those of Gaitan [23]. However, the raw values of the NTL median showed a highly skewed distribution (skewness = 3.97) and included zero values, making them inappropriate for direct logarithmic transformation. To address this issue, we applied a transformation to reduce skewness and ensure that all values were strictly positive. Specifically, we used the transformation

$$LogNTL = log(median + K), \tag{1}$$

where $K = 26.68$. This optimization was achieved using a brute-force search for possible K values. This resulted in a distribution with minimal skewness. In our case, the final transformed variable had a skewness of approximately zero, indicating a nearly normal distribution and eliminating any negative values. In our first approach, we use this log-transformed median NTL value as the seventh channel in our CNN input. That is, for each AGEB, all pixels in the corresponding region were assigned the same value: the logarithm of the AGEB's median nighttime light intensity. Then we appended these values to the patches obtained from the six-band channels of the Geomedian into a single numeric array.

For our second approach, we fuse day and night remote sensing imagery by combining the two-pixel numeric arrays into a single array. In addition to the first approach, we obtained 40 × 40 patches of the Geomedian. Then, since the resolution of both sources is not equal, we use three different resampling methods to downscale the resolution of NTL images to match the 30 m pixel resolution of Geomedian

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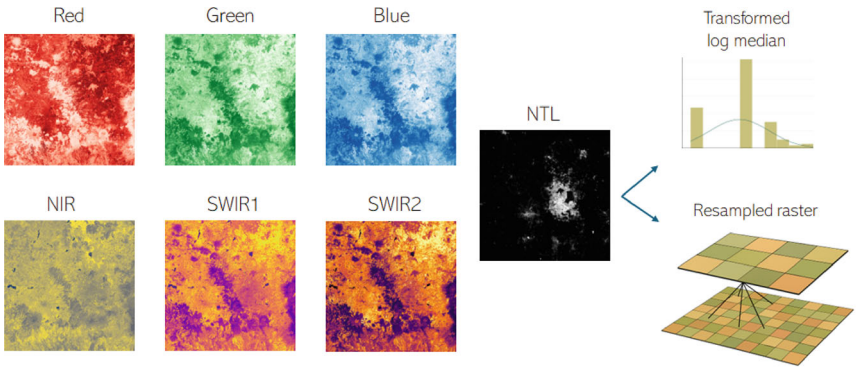

Fig. 1 Examples of Landsat-8 bands (RGB, SWIR1, SWIR2, NIR) and VIIRS NTL band of Mexico City for 2020 and the two proposed methods for NTL integration

images (see Fig. 1). As in Yeh et al. [10], we up-sampled VIIRS images using the Nearest-Neighbor algorithm since it does not modify pixel values. We also use bilinear (2×2 kernel) and cubic (4×4 kernel) interpolation methods to reduce the blocky or stepped appearance introduced by nearest-neighbor resampling, and to test the impact of using smoother, though less radiometrically accurate images [24].

3.3 CNN Configuration and Validation Approach

We trained five CNN models with the same configuration, that is: data augmentation transformations (horizontal and vertical flip, grid distortion, elastic deformation and rotation), learning rate of 10^{-4}, Adam optimizer, initialized with the ImageNet trained model weights for the layers compatible with the 3-channel input EfficientNet-B3 architecture, and 30 epochs. The first five layers of the CNN models are not compatible with the pre-trained model; nevertheless, they do not use activation functions. We added at the top of the models a global average pooling layer, a dropout layer with a dropout rate of 0.5, and a dense layer with one-unit output and sigmoid activation [21]. We initialized this layer with the Xavier method [25], since the sigmoid function is differentiable, unlike ReLU or LeakyReLU [26]. L2 regularization ($\lambda = 0.1$) was applied to the convolutional layers. Training was performed using the mean squared error loss, with early stopping and learning rate reduction implemented when the validation loss plateaued. Our final data split consisted of 50% training, 20% for validation, and 30% for testing data.

To test if our methods improve the poverty prediction performance of the models, we predict poverty using only a Geomedian six-band array, which proved to be a reliable approach for poverty inference at more granular levels [21]. We also weighted averages at municipal and state levels using population data used by CONEVAL to compare the results with the municipal and state multidimensional

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poverty rate reported by CONEVAL for 2020 and determine the precision of our models in years where granular population data are not available.

4 Results

Remote sensing for mapping and predicting socioeconomic indicators has been widely tested and proven to be a reliable, inexpensive, and effortless approach. In this study, we explored and evaluated two methods for combining daytime and nighttime satellite images for poverty prediction at AGEB level using CONEVAL municipal multidimensional poverty rate by using CNN regression models. We use the coefficient of determination (R^2) to evaluate the effectiveness of poverty prediction models [27]. We employ population data from CONEVAL and the 2020 Census to form weighted averages for comparison with CONEVAL's official multidimensional poverty statistics. Our aim is to determine the most accurate fit at the municipal level using a model developed at the AGEB level, where variations specific to AGEBs are accounted for at the municipal tier through CNN networks (Table 2).

Unlike previous studies that achieved R^2 values between 0.55 and 0.84 at different spatial scales using NTL and CNN models [4–11, 14, 18], our results show that the combination of Landsat-8 bands with VIIRS NTL not only outperforms the six-band baseline model ($R^2 = 0.66$ at the municipal level), but also achieves performances comparable to those reported in the literature. The rescaled seven-band model reached an R^2 of 0.73 at the municipal level and 0.96 at the state level, showing that, even when using municipal data as a proxy variable, the integration of daytime and nighttime information improves poverty prediction at both municipal and state levels.

Due to the low resolution of VIIRS images, we find that there is no significant difference between resampling methods. However, the three resampling methods used in this study performed as well as the transformed log-median approach. Our

Table 2 Results of CNN regression models

Model	**NTL** integration	$\mathbf{R^2}$ at municipal level	$\mathbf{R^2}$ at state level
Landsat-8 6 spectral bands	None	0.66	0.92
Landsat-8 + VIIRS NTL log median	Transformed log median	0.71	0.94
Landsat-8 + rescaled VIIRS (1)	Resampled image by NN method	0.76	0.96
Landsat-8 + rescaled VIIRS (2)	Resampled image by bilinear method	0.71	0.95
Landsat-8 + rescaled VIIRS (3)	Resampled image by cubic method	0.73	0.96

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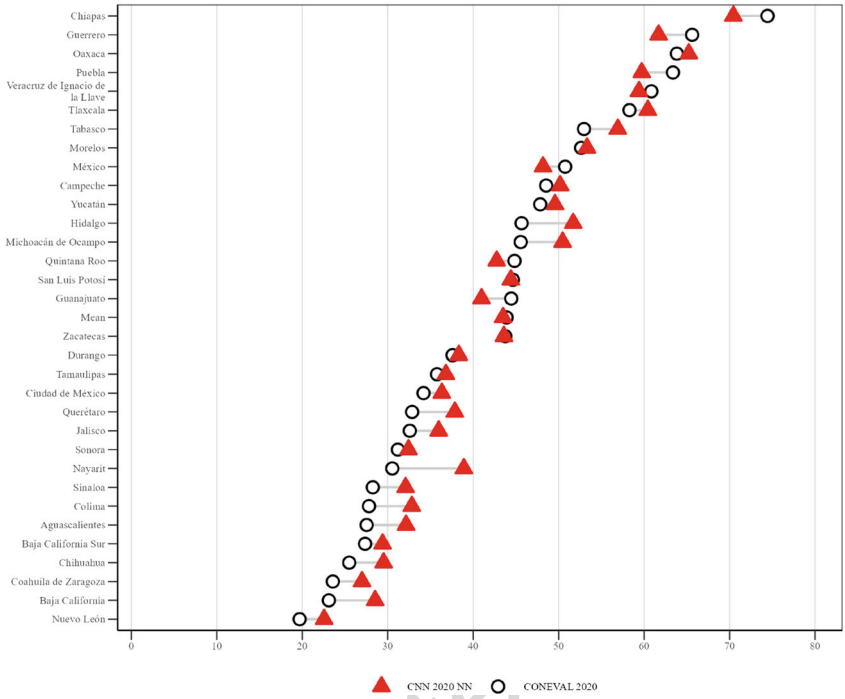

Fig. 2 Comparison of CONEVAL 2020 and CNN (NN) multidimensional poverty estimates across all mexican states. Note: state-level estimates produced with the CNN model using Landsat-8 and rescaled VIIRS by NN method

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findings demonstrate that combining Landsat-8 images with the VIIRS NTL annual composite in a single numeric array with the two presented approaches for CNN regression models improves poverty prediction at three different aggregation levels in Mexico.

Figure 2 shows survey estimates improve with CNN methods, achieving high predictive power at the state level. Because our model predicted 2020 results accurately, we plan to test its ability to predict 2024 poverty using the same architecture.

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5 Conclusions and Future Work

In future research, we might consider implementing a multiscale attentional model, which could potentially enhance effectiveness. Importantly, Nearest Neighborhood upsampling has shown satisfactory outcomes at the municipal level, whereas cubic upsampling proved more effective at the AGEB scale. Future exploration could involve investigating various visual characteristics to enhance our existing approach.

Moreover, a multiscale attentional model may better focus on these visual cues. An attention-based transformer method could be even more effective in highlighting distinguishable visual features that indicate poverty or in identifying features captured by Landsat or NTL that might contribute to poverty alleviation. However, training such a model would incur higher costs compared to EfficientNet.

Researchers continuously modify the architectures of deep learning models to improve their performance in regression and classification tasks. For the study of poverty using remote sensing, where visual features exist at different resolutions, the Multiscale Vision Transformer (MViT) is a particularly relevant model to consider for future research. Moreover, in future studies, we can use CNN models designed in 2020 to forecast poverty rates for the year 2024. This goal can be achieved by deploying updated satellite data specific to that year, using architectures that have been verified earlier. The findings in this document have significant potential to benefit policymakers by comparing poverty estimates from various sources, particularly in cases where prolonged data generation processes, such as extensive surveys and censuses, require substantial financial investment. The information generated by previously trained CNN models can provide timely information to assess poverty evolution to assess social policy effectiveness. The development of this method is significant for strategizing solutions to pressing challenges, including water scarcity, climate change, and changes in land use. The underlying principles are adaptable and can be applied within these areas to implement socio-environmental policies.

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Author Contribution All authors contributed equally.

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AQ2. Missing citation for Tables "1 and 2" were inserted in text. Please check if appropriate. Otherwise, please provide citation for Table "1 and 2". Note that the order of main citations of tables in the text must be sequential.

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